**Title : Project Registration & Progress Review** **FF No. 180**

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| **Department:** Computer Engineering | **Academic Year:** 2026-27 |
| **Semester :** SEM- V | **Group No. : TY-A2** |
| **Project Title:** Explainable Feature Fusion and Ensemble Learning for Modi Script Recognition | |
| **Project Area:** Artificial Intelligence and Data Science | |

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**FF No 180**

**Project Synopsis**

## 1. Introduction

Explainable Feature Fusion and Ensemble Learning for Modi Script Recognition: is a recognition system for handwritten Modi script. Modi is the cursive script used to write Marathi for about 700 years in land records, court documents and administrative correspondence across Maharashtra. Very few people can read it today, so millions of archived documents are effectively unreadable. The system deliberately uses classical machine learning with handcrafted features instead of deep learning: three complementary descriptors, Histogram of Oriented Gradients (HOG), Zernike moments and Local Binary Patterns (LBP), are fused and classified by Random Forest, SVM (RBF), XGBoost and a stacking ensemble, and every prediction is explained with SHAP. The work is delivered as an interactive web application that recognises the characters in an uploaded image and shows why each decision was made.

## 2. Problem Statement

Millions of historical Modi documents cannot be read because very few experts remain who know the script. Earlier automatic recognition studies do not solve this reliably: deep-learning studies report 86-98.92% accuracy on a benchmark of only 90 images per class and show clear signs of overfitting, while classical studies report 97.77-99.3% on private datasets of undisclosed size, so their results cannot be verified. None of the surveyed studies explains its predictions or analyses which characters are confused with each other. An accurate, reproducible and explainable recognition system evaluated on a public dataset is therefore needed, together with a measurement of the training-set size at which handcrafted features stop being competitive with deep learning.

## 3. Objective and Scope

Objectives

- To recognise all 57 handwritten Modi character classes (12 vowels, 35 consonants and 10 digits) using handcrafted features and classical machine learning instead of a deep network. Most recent papers, such as Bhong and Kulkarni (2025) and Kolte and Gupta (2025), train CNNs on about 90 images per class, and their own numbers show the models overfitting.
- To combine three kinds of features, HOG for stroke direction, Zernike moments for overall shape and LBP for small local patterns, and to check through an ablation study how much each one adds. Kulkarni et al. (2014) tried Zernike moments on just 100 digit images, and Mandke et al. (2024) gave their SVM and KNN raw pixels, so a proper fused feature set has not been tried on Modi yet.
- To train Random Forest, SVM and XGBoost on these features, compare them on the same test set, and see whether stacking the three beats the best single model. Ekbote et al. (2022) also used Random Forest and XGBoost, but on VGG16 features and one model at a time.
- To explain every prediction with SHAP, and to find out which character pairs get confused the most and which part of the character causes the confusion. Anam and Gupta (2015) and Kolte and Gupta (2025) both point out that similar looking characters are the main problem, but neither of them lists which ones.

- To find the training set size at which a plain CNN starts doing better than our classical pipeline. Both are trained on the same splits, from 72 to 2,000 images per class, and tested on the same 11,400 images, so the comparison is fair. Chandankhede et al. (2025) got 97% on test but only 45% on validation with about 21 images per class, which is why we wanted to measure this instead of just assuming deep learning wins.
- To make a web app that works on both phones and laptops, where you can upload a photo or draw a character and see the predicted character, how confident the model is and why it decided that. It should also split a line of separate characters, since Solanki and Ingle (2019) only worked with single characters.

Scope

- Recognition of 57 isolated handwritten Modi character classes (10 digits, 12 vowels and 35 consonants) from the public MODI-HChar dataset of 575,912 images.
- Image preprocessing: grayscale conversion, Otsu binarisation, scan-frame and noise removal, and size normalisation.
- Segmentation of images containing several separated characters into individual characters in reading order.
- Feature extraction, dimensionality reduction (PCA / LDA), classification and per-class error analysis.
- Explainable output for every prediction through a web application that works on phones and laptops.
- Reliable segmentation of continuous cursive manuscripts and full document transliteration are kept as future scope.

## 4. Methodology

The methodology follows a reproducible classical machine-learning pipeline with a fixed random seed:

- Data Collection
  - Download the MODI-HDC dataset from Kaggle and count the classes and images from disk instead of relying on published figures.
  - Create one seeded ordering per class: a fixed test set of 200 images per class and nested training sets of 72 to 2,000 images per class.
- Data Pre-processing
  - Convert to grayscale, apply median filtering and Otsu binarisation with automatic ink polarity detection.
  - Remove scan-frame lines and specks, reject blank images, and centre and resize each character to 64 x 64 pixels.
- Segmentation of Multi-character Images
  - Correct skew and find text lines with the horizontal projection profile.

  - Split characters using the vertical projection, headline (shirorekha) handling and connected components, and flag unreliable boxes.
- Feature Extraction
  - HOG for stroke direction (1,764 features) and Zernike moments for rotation-invariant global shape (49 features).
  - LBP for local stroke patterns at junctions and stroke ends (448 features).
- Feature Fusion and Dimensionality Reduction
  - Standardise and fuse the three descriptors with family-balanced weighting (2,261 features).
  - Compare PCA and LDA using cross-validation on the training data only, and keep the better one (LDA).
- Classification
  - Train Random Forest, SVM (RBF) and XGBoost separately and combine them in a stacking ensemble with a logistic-regression meta-learner.
  - Use stratified splits and cross-validation on the training data only.
- Evaluation
  - Report accuracy, macro precision, recall and F1, per-class and per-category results, the confusion matrix and the most confused character pairs.
  - Run an ablation of single descriptors against fused features and a learning curve against an untuned reference CNN on identical splits.
- Explainability and Deployment
  - Explain predictions with grouped SHAP: global importance, per-prediction explanations and contrastive explanations for confused pairs.
  - Deploy a web application with Home, Read, Results, Explain and Script pages.

**5. Hardware & Software**

**Hardware Requirements**

- Processor: Intel Core i5 / AMD Ryzen 5 or above (CPU only; no GPU required)
- RAM: Minimum 8 GB (16 GB recommended for training)
- Storage: Minimum 20 GB of free space (dataset download and cached features)
- Internet Connection: Required only for the one-time dataset download

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python 3.11 or above
- Image Processing: OpenCV, scikit-image, mahotas
- Machine Learning: scikit-learn, XGBoost
- Explainability: SHAP
- Data Processing and Visualisation: NumPy, Pandas, Matplotlib
- Reference Baseline only: PyTorch (CPU)
- Application Development: Starlette (Python web framework) with HTML, CSS and JavaScript
- Development Environment: Jupyter Notebook / Visual Studio Code, Git

## 6. Environment

The project is designed to have a low environmental footprint. It runs entirely on an ordinary laptop CPU without GPUs, uses classical models that train far faster than deep networks, and caches intermediate results such as processed images and extracted features so that experiments are not recomputed. Digitising and reading archival documents also reduces the need to physically handle fragile paper records, which helps to preserve them.

## 7. Sustainability

- Sustainability covers both technical sustainability and long-term usefulness of the system.
- Project-specific considerations: The system is designed to be:
- Modular - preprocessing, segmentation, feature extraction, models, evaluation and explanation are separate modules that can be improved independently.
- Reproducible - a fixed random seed, pinned library versions and cached results give the same output on every run.
- Configurable - every tunable parameter is kept in one configuration file.
- Lightweight - CPU-only training and inference make it usable without specialised hardware.
- Extensible - new descriptors, classifiers, character classes or datasets can be added without redesigning the pipeline.
- Tested - automated tests check the data splits, preprocessing, features, segmentation and the web application.

## 8. Safety

The system is a research prototype that assists human readers of Modi documents; it is not an authoritative transcription service. Its safety measures focus on making uncertainty visible and on secure handling of uploaded images.

### Safety Considerations

- Confidence score for every predicted character with green, amber and red levels.
- Clear flags for unreliable segmentation (merged, fragmented or connected characters).
- Top-3 alternatives and SHAP explanations so that users can check every decision.
- Honest reporting of accuracy limits and failure cases.
- Human verification of transcriptions before any scholarly or legal use.

### Cybersecurity

- Uploaded images are processed locally and are not stored or shared.
- Validation of uploaded files (type, size and readability) before processing.
- Upload size limit to prevent misuse of the application.
- Dataset API credentials are kept out of version control.

## 9. Ethics

Modi documents are part of Maharashtra's cultural heritage and may contain land, legal and family records. The project addresses the responsible use of AI for such material through transparency, honest reporting and human oversight.

**Important Considerations**

- Transparency: Every prediction is accompanied by a SHAP explanation showing which parts of the character influenced the decision.
- Honest Reporting: Results are reported on a fixed public test set, including cases where a method performed worse than expected.
- Data Licensing: The MODI-HDC dataset is published under a CC0 public-domain licence.
- Dataset Bias: Handwriting from a limited group of writers may not represent every historical period or writing style; this limitation is stated clearly.
- Human Oversight: Outputs are aids for trained readers and are not treated as authoritative transcriptions.
- Cultural Respect: Character labels and transliterations are checked against reference examples instead of being guessed.

**10. Cost**

The project has a very low cost because it uses a free public dataset, open-source software and an ordinary laptop; no GPU, paid API or cloud training is required.

**Cost Factors**

- Dataset acquisition: free (public Kaggle dataset, CC0 licence)
- Software and libraries: free and open source
- Hardware: existing laptop; no GPU or cloud training needed
- Development time for implementation, experiments and documentation
- Optional cloud hosting of the web application for demonstration
- Maintenance and retraining on new data

**Potential Economic Benefits**
**The system can potentially reduce:**

- Manual effort and time needed to read archival Modi documents
- Dependence on the very small number of expert Modi readers
- Cost of digitising and indexing historical archives
- Effort needed by students and researchers to learn the script

**11. SDG Project Mapping**

SDG 4 - Quality Education

- Provides an interactive tool for learning to read Modi characters.
- Explains each prediction, helping learners see how similar characters differ.
- Supports students and researchers who work with historical sources.

SDG 9 - Industry, Innovation and Infrastructure

- Applies data science and machine learning to document digitisation.
- Develops an explainable, CPU-friendly recognition system.
- Contributes an open, reproducible pipeline that others can build on.

SDG 11 - Sustainable Cities and Communities (Target 11.4: cultural heritage)

- Helps preserve and give access to Maharashtra's documentary heritage.
- Supports archives in digitising fragile historical records.
- Makes historical land and administrative records easier to read.

**12. Application and Future Scope**

Application

- Archive Digitisation: Assist archives and libraries in reading and indexing handwritten Modi documents.
- Historical Research: Help historians read land records, court documents and correspondence.
- Education: Serve as an interactive learning aid for students of the Modi script.
- Assisted Transcription: Provide character-level suggestions with confidence scores for expert verification.
- Explainable AI Demonstration: Show how handcrafted features and ensemble models reach their decisions.

Future Scope

- Reliable segmentation of continuous cursive manuscripts, including touching characters.
- Word-level and line-level recognition with transliteration into Devanagari.
- Recognition of vowel signs (matras) and conjunct characters beyond the 57 base classes.
- Training on more writers and historical eras to improve generalisation.
- Hybrid models that combine handcrafted features with learned features.
- A mobile application for on-site use in archives.

| Group No. | TY-A2 | | |
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| Activity | Review Schedule | Progress Review Report submitted | Signature of Guide |
| Review 1 | Mid Sem. Semester | Yes / No | |
| Review 2 | End of Semester | Yes / No | |

Format of Progress Review Report:

| **Review No.: 1** | **Group No.: TY-A2** | **Date:** |
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| **Progress Review Report** | | |
| **Signature of Guide:** | | |

| **Review No.: 2** | **Group No.: TY-A2** | **Date:** |
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**Progress Review Report**



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